

Each question is worth 5 points for a total of 40 points. The Extra Credit problem can get you up to an additional 5 points.

Problem 1

Solve $\int \sqrt{3 - 2s} \, ds$.

Problem 2

Solve $\int \frac{9r^2}{\sqrt{1-r^3}} \, dr$.

Problem 3

Solve $\int_{-3}^3 \frac{3}{(x+3)^4} dx$.

Problem 4

Solve $\int_0^{\pi/4} \tan^2(x) dx$. Hint: use your trig identities to rewrite $\tan^2(x)$.

Problem 5

If $y = \int_0^{x^2} \sqrt{1+t^2} dt$, find $\frac{dy}{dx}$. Hint: let $u = x^2$ and use the chain rule.

Problem 6

Solve $\int \sec(2x) \tan(2x) dx$.

Problem 7

Use the definition of a definite integral to solve $\int_0^b \frac{x}{2} dx, b > 0$. Note: you will need your summation formulas.

Problem 8

Use the definition of a definite integral to solve $\int_a^b x^2 dx$. Note: you will need your summation formulas.

Extra Credit:

The velocity of a particle moving back and forth on a line is $v = \frac{ds}{dt} = 2t$ m/sec for all t . If $s = 0$ when $t = 0$, find the value of s when $t = \frac{\pi}{2}$ sec.